

The final echelon form: $\begin{bmatrix} \square & * & * & * \\ 0 & \blacksquare & 0 & \square & * \\ 0 & 0 & 0 & \square \end{bmatrix}$

- By condition 2 from the defn. of echelon form, we know that Row 1 cannot have its pivot in Column 3 and Column 4 because otherwise, A will have 2 pivots and 1 pivot respectively which is a contradiction.
- $\therefore$ Row 1 Column 2 = $\square$. From defn. of pivot, Row 1 Column 1 = 0.
- It's easy to show that by Condition 2 from the defn. of echelon form, Row 2 will have its pivot in Column 3 and not Column 4, since 0.w. A will have only 2 pivots which is a contradiction.
- $\therefore$ Row 2 Column 3 = $\square$. From defn. of pivot, Row 2 Column 1 = Row 2 Column 2 = 0.
- $\therefore$ Row 3 Column 4 = $\square$. From defn. of pivot, Row 3 Column 1 = Row 3 Column 2 = Row 3 Column 3 = 0.
- $\therefore$ The final echelon form: $\begin{bmatrix} \square & * & * & * \\ 0 & \blacksquare & 0 & \square & * \\ 0 & 0 & 0 & \square \end{bmatrix} \square$

31. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, with A its standard matrix. Complete the following statement to make it true: "T is one-to-one if and only if A has _____ pivot columns." Explain why the statement is True.

Solution: From Theorem 10, A is a $m \times n$ matrix. By Theorem 11, T is one-to-one if and only if the equation $T(x) = 0$ has only the trivial solution. $\therefore Ax = 0$ has only the trivial solution.

From Theorem 3, the matrix equation $Ax = 0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

- The augmented matrix $[A \ 0]$ can be converted to a echelon form using a Row Reduction algorithm.
- Since the solution is unique, By Theorem 2, there are no free variables in the system.